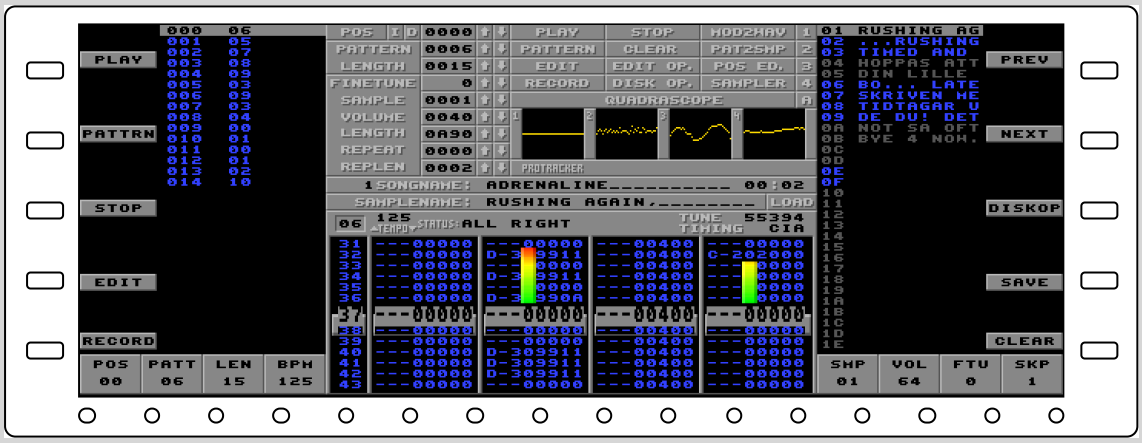
TRACKER

ProTracker 2.3D

for the KN7000

Operating Instructions



Homebrew software for the Technics SX-KN7000; not a Technics or Panasonic publication.
Edition of 28 October 2026, for the program in kn5000_homebrew/demos/kn7000_protracker (the folder PROTRACK\ on the SD card). A port of pt2-clone by Olav Sørensen (8bitbubsy), the faithful C version of ProTracker 2.3D (BSD 3-Clause). The KN7000 side and this manual: GPLv3 or later. Headings set in Archivo (SIL Open Font License 1.1).

Contents

PART I GETTING STARTED

Introduction 3
What goes on the SD card 3
Starting and leaving 4
The screen 5
Controls and functions 6

PART II PLAYING

Playing modules 7

PART III EDITING

Editing a pattern 8
Changing what is there 9
Arranging the song 10
Samples 10

PART IV FILES

Loading and saving 11

PART V PRACTICAL APPLICATIONS

A first session, step by step 12

PART VI REFERENCE

Every control 13
ProTracker's effects 14
What differs from the Amiga 15
Troubleshooting 15

HOW THIS MANUAL WRITES THINGS

- Buttons and controls of the panel are printed in bold capitals, as on the panel: **START/STOP**, **SOUND GROUP**, **TEMPO/PROGRAM**.
- **LCDL 1-5** and **LCDR 1-5** are the five buttons left and right of the display, counted from the top. **MUTE UP 1-16** and **MUTE DOWN 1-16** are the pairs under it, counted from the left.
- **SHIFT** means holding **DISPLAY HOLD** while pressing the other control.
- Words and messages on the display are printed as the display shows them: MODULE SAVED !, POS, SV.NEW.

Introduction

What this is

ProTracker is the Amiga's classic music tracker. A song (a *module*, a .MOD file) is a list of *positions*. Each position plays a *pattern* of 64 rows, and each row has four channels. A cell of a channel holds a note, the number of the sample it plays (1-31), and an effect: a command digit and a two-digit parameter, all in hexadecimal. The cursor moves through the cells, and the keyboard enters notes and digits. On the Amiga that keyboard is a computer keyboard and a mouse.

On the KN7000 the same program runs on the instrument. Its replayer plays the four channels through the instrument's effects processor (DSP), and its screen is drawn on the display. The KN7000 has no computer keyboard and no mouse, so **the panel and the keybed do their work**. The keybed plays and enters notes. Each button was chosen so that what it does on the KN7000 suggests what it does in ProTracker: the sequencer's buttons are the song, the **SOUND GROUP** is a keypad of hex digits, **PANEL MEMORY** stores and recalls effects, and so on (see "Controls and functions").

Every module with four channels and 31 samples that the Amiga's ProTracker makes or plays works: ProTracker, NoiseTracker and Soundtracker modules. The examples in this manual use the music of *Pinball Fantasies*.

What goes on the SD card

The folder PROTRACK\ at the card's root holds the program and your modules:

file	what it is
KN7APP.BIN, PT.BIN	the program
MANIFEST.TXT	the name the program menu shows: "ProTracker 2.3D"
.MOD, MOD.	the modules: TABLE1.MOD, PTNEW01.MOD, or Amiga-style names such as MOD.JAZ

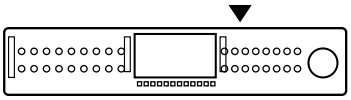
■ Adding modules

Copy a module into PROTRACK\ from a computer. The program lists every file there whose name ends in .MOD or starts with MOD., in order of name, and steps through them in that order.

■ Long file names

A file with a long name is listed by the short 8.3 name the computer gave it on the card: Space_Debris.mod shows as SPACE_~1.MOD.

Starting and leaving



Starting ProTracker

1 Insert the KN7LOAD floppy disk and the SD card.

2 Press the **DISK MENU** button, then **STYLE CONVERT** (left button 3).

3 Choose KN7LOAD.CNV and press **OK**.

- The program menu appears.

4 Select "ProTracker 2.3D" with the left buttons and press **RUN**.

5 The first module, by name, loads and plays.

- Its samples are sent to the DSP first. This takes between half a second and two seconds, and the status line says SENDING SAMPLES...

Leaving

1 Hold **EXIT** for 1.5 seconds.

- The instrument shows its message, then "Continue Next?".

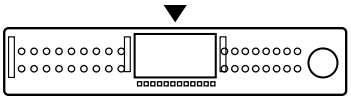
2 Answer **NO** (right button 4) to go back to the instrument.

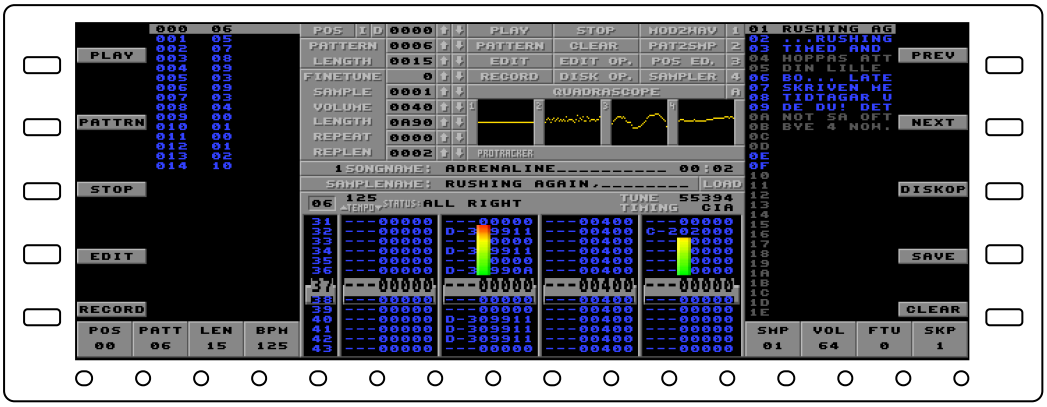
- A brief press of **EXIT** is ProTracker's Esc instead: it goes back from a page, the macros or the edit mode.

■ Save before leaving

Edits live in the instrument's memory only until you save them (see "Loading and saving"). Leaving, or loading another module, loses unsaved edits; loading asks first.

The screen



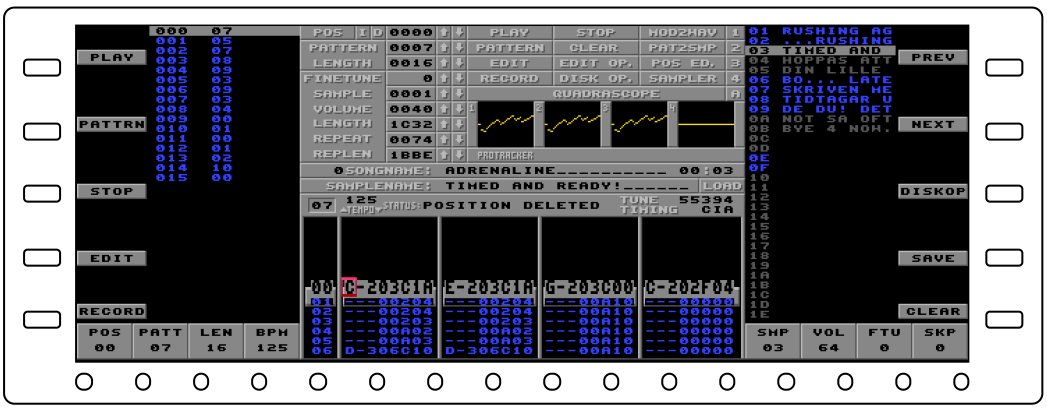


While a module plays.

The display is 640 dots wide and ProTracker's screen 320, so ProTracker sits in the middle, as it looks on the Amiga. Its pattern shows 13 rows instead of 15: the display is 240 lines high, the Amiga's screen 255. The 160 dots on either side show what ProTracker hides behind other screens on the Amiga:

- **The soft keys' tags.** Beside each of the five buttons left and right of the display, the word for what it does now. They change with the page (main, Disk Op., options) and become YES and NO during a question.
- **The positions (left).** The song's positions and the pattern each plays, the current one boxed: ProTracker's position editor, always in view. **BANK VIEW** shows the effect macros here instead.
- **The samples (right).** All 31 sample names, the current one boxed; empty samples dimmed. In the Disk Op. this column lists the modules instead.
- **The values (bottom).** Four boxes on each side, each above one of the 16 **MUTE UP**/**DOWN** pairs under the display. The four leftmost pairs and the four rightmost change the value shown right above them.

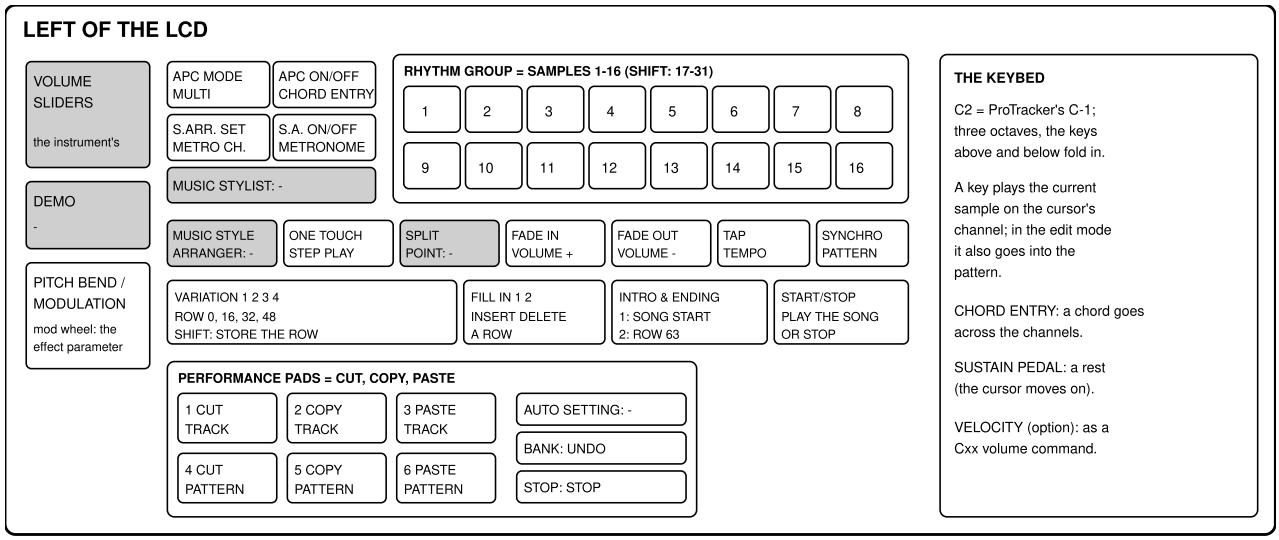
ProTracker's own screen works as on the Amiga: POS, PATTERN and LENGTH at the top left, the current sample's number, volume, length and loop below them, the quadrascope (or spectrum analyzer) at the top right, the song's and the sample's names, the status line, the tempo (BPM), and the pattern with its VU meters. Its buttons (PLAY, DISK OP. and the rest) are drawn but not pressed: the panel does that.



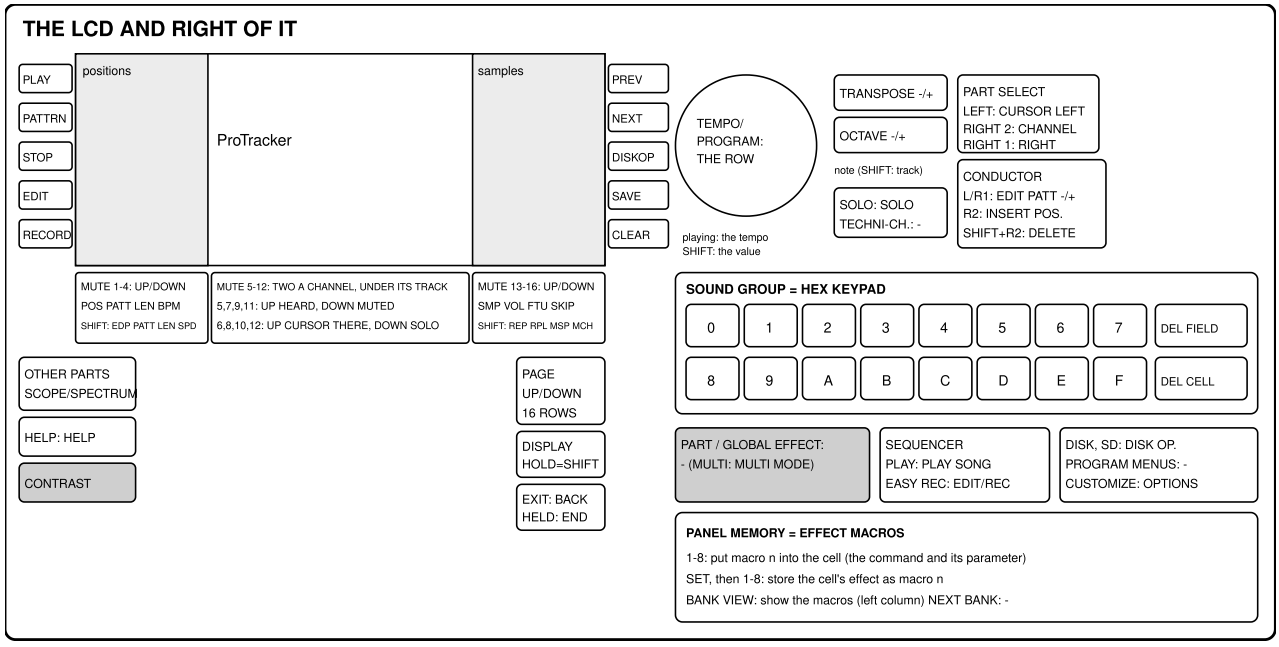
The edit mode: the red cursor on the middle row, on channel 1's note. A chord from the keybed went into row 0 across channels 1-3 (C-2, E-2, G-2, sample 03).

Controls and functions

Each section of the panel takes one of ProTracker's jobs. The two drawings show the left and the right halves of the panel. The controls drawn in grey are left to the instrument; "-" marks a button that does nothing in ProTracker.



The left half of the panel and the keybed.



The display, its soft keys and the **MUTE** pairs under it, and the right half of the panel.

■ SHIFT

Hold **DISPLAY HOLD** and press another control for its second function. The **TEMPO/PROGRAM** wheel's LED blinks while it is held, and the values at the bottom of the display change to show what the **MUTE** pairs do with SHIFT.

Playing modules

The song, the pattern, the channels

to	press	ProTracker's
play the song from the current position	**START/STOP**, **SEQUENCER PLAY** or **LCDL 1**	right Amiga key
play the current pattern over and over	**SYNCHRO & BREAK** or **LCDL 2**	right Alt
stop	**START/STOP**, **PADS STOP** or **LCDL 3**	space
the next or the previous module	**LCDR 2**, **LCDR 1**	Disk Op.
change the tempo while playing	turn **TEMPO/PROGRAM**, or tap **TAP TEMPO** four times on the beat	the BPM arrows
go to another position while playing	**MUTE UP 1** / **MUTE DOWN 1** (POS: up or down the list)	the POS arrows
mute or hear a channel	**MUTE DOWN** / **MUTE UP** 5, 7, 9, 11 (under channels 1-4)	the mute buttons
solo a channel	**MUTE DOWN** 6, 8, 10, 12, or **SOLO** for the cursor's channel	right-click a mute button
the quadrascope or the spectrum analyzer	**OTHER PARTS/TG**	click the visualizer
the volume of the current sample	**FADE IN** / **FADE OUT**	the VOLUME arrows

■ The keybed

The keybed always plays: a key sounds the current sample on the cursor's channel, at its note. While a song plays, that channel's own notes take over again at its next note. **C2** on the keybed is ProTracker's C-1; three octaves are used, and the keys above and below fold into them.

■ The metronome

SOUND ARRANGER OFF/ON turns ProTracker's metronome on and off, and **SOUND ARRANGER SET** puts it on the cursor's channel. It plays sample 1F every few rows (SHIFT + **MUTE 15** sets how many).

■ Volume

The instrument's own sliders set the loudness of everything, as they do for the instrument's own sounds.

Editing a pattern

The edit mode

Press **EASY REC** (or **LCDL 4**). Playing stops, and the red cursor appears on the middle row. Press **EASY REC** again, or **EXIT** briefly, to leave the edit mode.

While a pattern is playing, **EASY REC** starts the *record* mode instead (**LCDL 5** too): the pattern plays over and over and what you enter goes into it at the row being played.

Moving the cursor

to move	press
a row up or down	turn **TEMPO/PROGRAM**
16 rows	**PAGE UP**, **PAGE DOWN**
to row 0, 16, 32, 48	**VARIATION 1-4** (SHIFT + **VARIATION** stores the current row in that button)
to row 63	**INTRO & ENDING 2**
a column left or right	**PART SELECT LEFT**, **PART SELECT RIGHT 1**
to the next channel	**PART SELECT RIGHT 2**
to channel 1-4	**MUTE UP** 6, 8, 10, 12 (the pair under that channel)

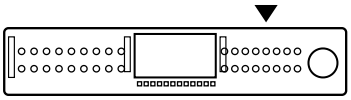
Each channel has six columns, as in ProTracker: the note, the sample number's two digits, the effect's command and its parameter's two digits.

Entering

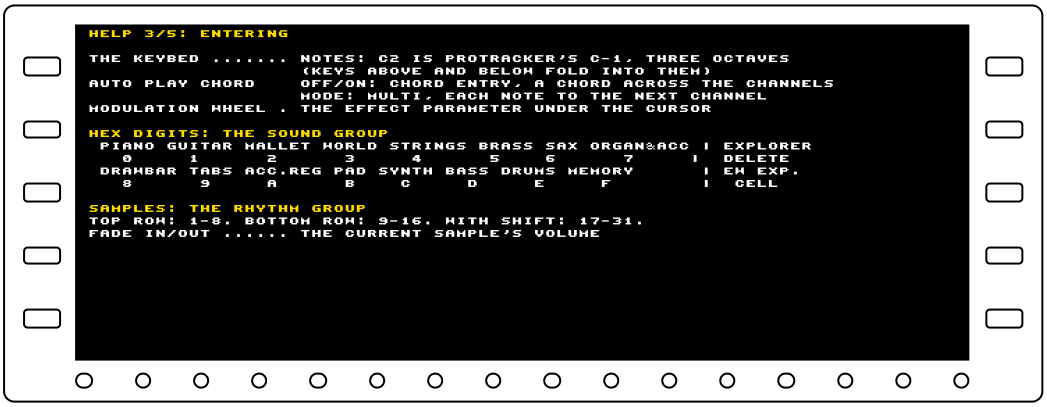
- **Notes:** play the keybed. The note and the current sample's number go into the cell under the cursor, and the cursor moves down by the *edit skip* (SKP; **MUTE 16** changes it, 0 to 9).
- **Digits:** on a sample or effect column, press the **SOUND GROUP** button for the digit: the top row is 0-7, the bottom row 8-F. The cursor moves down by the edit skip, as ProTracker's does. Set the skip to 0 to type a whole effect in one row.
- **Chords:** turn **AUTO PLAY CHORD OFF/ON** on and play a chord: its notes go into the cursor's channel and the ones to its right, on the same row.
- **MULTI: AUTO PLAY CHORD MODE** or **MULTI**: each note you play goes into the next channel, so a melody can be spread over the channels as you play.
- **Rests:** the sustain pedal moves the cursor on without a note.
- **The modulation wheel** sets the effect parameter under the cursor (when the cursor is on one), 00 to FE: handy for volumes (Cxx) and slides.
- **SHIFT + TEMPO/PROGRAM** changes the value under the cursor a step at a time: the note a semitone, a digit by one.

(Continued on the next page)

Changing what is there



to	press	ProTracker's
delete the note (on the note column) or the effect (elsewhere)	**SOUND EXPLORER**	Del
delete the whole cell	**EW EXPANSION** (or SHIFT + **SOUND EXPLORER**)	Shift+Del
insert an empty row at the cursor (the track's rows below move down)	**FILL IN 1** (SHIFT: all four tracks)	Shift+Return
delete the row at the cursor (the rows below move up)	**FILL IN 2** (SHIFT: all four tracks)	Backspace
transpose the note a semitone	**TRANSPOSE -** / **+** (SHIFT: the whole track)	Edit Op. 1
transpose the note an octave	**R1**/**R2 OCTAVE -** / **+** (SHIFT: the whole track)	Edit Op. 1
cut, copy, paste the track	**PAD 1**, **PAD 2**, **PAD 3**	Shift+F3, F4, F5
cut, copy, paste the pattern	**PAD 4**, **PAD 5**, **PAD 6**	Alt+F3, F4, F5
undo the last change	**PADS BANK** (again: redo)	(pt2-clone's undo)
put an effect macro into the cell	**PANEL MEMORY 1-8**	Alt+1-0
store the cell's effect as a macro	**SET**, then **PANEL MEMORY 1-8**	Ctrl+1-0
see the macros	**BANK VIEW**	
play one row and move on	**ONE TOUCH PLAY**	Return



HELP, page 3: the **SOUND GROUP**'s digits. **HELP** shows five pages of these reminders; **PAGE UP**/**DOWN** or the wheel turn them.

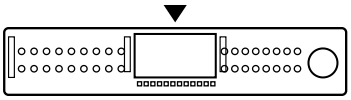
■ Effect macros

A macro is an effect command and its parameter. **PANEL MEMORY 1-8** put macro 1-8 into the cell under the cursor, as the instrument's PANEL MEMORY recalls a registration. Press **SET** and then a number to store the cell's effect there, as the instrument stores a registration.

■ Undo

PADS BANK takes back the last change to the pattern; pressing it again puts the change back.

Arranging the song



The song is the list of positions shown in the display's left column. Each line is a position and the pattern it plays.

to	press
go to another position	**MUTE UP 1** / **MUTE DOWN 1** (POS: the one above or below in the list)
change the pattern this position plays	**MUTE UP 2** / **MUTE DOWN 2** (PATT)
make the song longer or shorter	**MUTE UP 3** / **MUTE DOWN 3** (LEN)
insert a position (a copy of this one, after it)	**CONDUCTOR RIGHT 2**
delete this position	SHIFT + **CONDUCTOR RIGHT 2**
edit another pattern without changing the song	**CONDUCTOR LEFT** / **CONDUCTOR RIGHT 1** (the pattern number at the screen's bottom left)
go to the song's start	**INTRO & ENDING 1**
the tempo	**MUTE UP 4** / **MUTE DOWN 4** (BPM), **TAP TEMPO**, or the wheel while playing

■ The tempo is in the song

As in ProTracker, the tempo and the speed are not stored in the module's header: a song sets them with the F command in its first row (for example F06 for speed 6, F7D for 125 BPM). To keep a tempo, enter it as an Fxx effect.

Samples

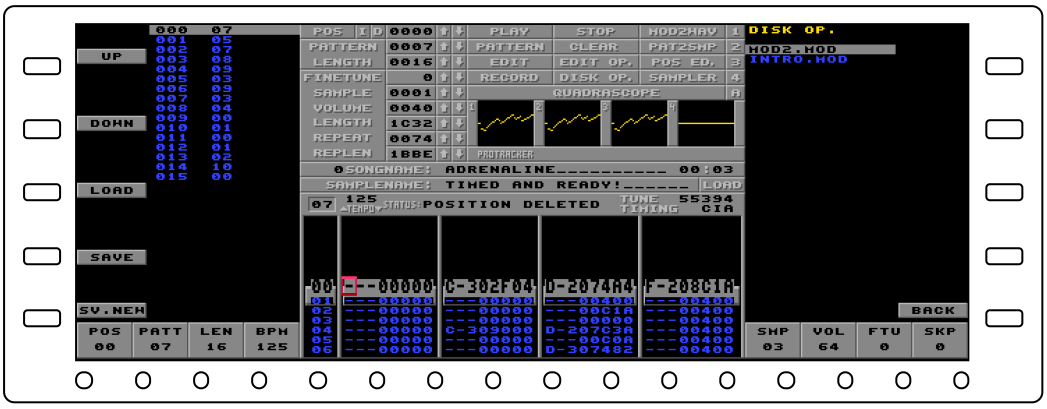
The current sample is the one notes play and enter. Choose it with the **RHYTHM GROUP**: its top row is samples 1-8 and its bottom row 9-16; with SHIFT, 17-31. Or step through them with **MUTE UP 13** / **MUTE DOWN 13** (SMP). The display's right column lists all 31.

value	pair	with SHIFT
SMP, the sample	**MUTE 13**	REP, the loop's start (hex, bytes)
VOL, the sample's volume (0-64); also **FADE IN** / **FADE OUT**	**MUTE 14**	RPL, the loop's length (hex, bytes)
FTU, its finetune (-8 to +7)	**MUTE 15**	MSP, the metronome's speed (rows a beat)
SKP, the edit skip (0-9)	**MUTE 16**	MCH, the metronome's channel (0: off)

- The samples' sounds themselves cannot be edited or recorded on the KN7000: there is no sample editor.

Loading and saving

The Disk Op.



The Disk Op. (**LCDR 3**, **DISK** or **SD**): the modules in the right column; the soft keys change to UP, DOWN, LOAD, SAVE, SV.NEW and BACK.

■ Loading a module

1 Press **LCDR 3** (DISKOP) to open the Disk Op.

2 Choose a module with the **TEMPO/PROGRAM** wheel (or UP, DOWN, **PAGE UP**, **PAGE DOWN**).

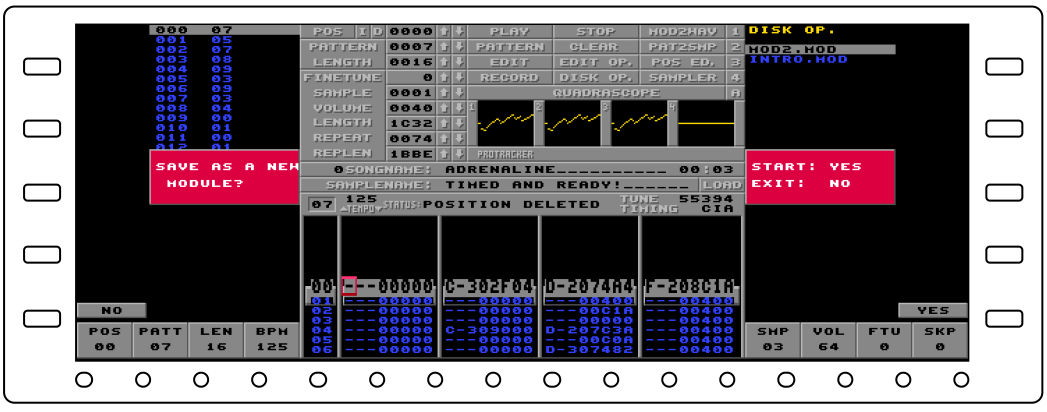
3 Press **LCDL 3** (LOAD).

- If the module in hand has unsaved edits, you are asked first.
- From the main page, **LCDR 1** and **LCDR 2** step through the modules without the Disk Op.

■ Saving

- **Save: LCDR 4** (or SAVE in the Disk Op.) writes the module over its own file, after asking.
- **Save as new:** SV.NEW (**LCDL 5** in the Disk Op.) writes PTNEW01.MOD, or the next free number. Your original stays as it was.
- **Clear: LCDR 5** empties the song (its patterns and positions), after asking; the samples stay.

Modules are saved as ProTracker 2.3D saves them ("M.K.", 31 samples), so they play in ProTracker, in every other tracker, and in pt2-clone on a computer.



A question. **START**/**STOP**, **SET** or **LCDR 5** (YES) answer yes; **EXIT** or **LCDL 5** (NO) answer no.

A first session

A four-bar beat, step by step

This makes a short beat from scratch inside a module, using its samples, and saves it as a new module.

1 Start ProTracker. The first module plays; press **START/STOP** to stop it.

2 Press **LCDR 3** (DISKOP), choose a module with the wheel, and press **LCDL 3** (LOAD). Press **START/STOP** to stop it again.

3 Choose an empty pattern to work in: press **CONDUCTOR RIGHT 1** until the bottom-left number shows a pattern the song does not use.

- The left column shows which ones it uses.

4 Press **EASY REC** to turn on the edit mode.

5 Choose the bass drum's sample in the **RHYTHM GROUP** (watch the right column), then press **VARIATION 1** (row 0). Set the edit skip to 4 with **MUTE UP 16**.

6 Play C3 on the keybed four times: a kick on rows 0, 4, 8 and 12.

- Keep going down the pattern with more notes, or with the sustain pedal for empty beats.

7 Press **PART SELECT RIGHT 2** (channel 2), choose the snare's sample, press **VARIATION 1**, and enter it on the off-beats.

8 Press **SYNCHRO & BREAK** to hear the pattern over and over. **START/STOP** stops it.

9 Add an effect. Back in the edit mode, set the edit skip to 0 (**MUTE DOWN 16**), move the cursor right three times to the command column, and press **SYNTH** (C), **PART SELECT RIGHT 1**, **STRINGS & VOCAL** (4), **PART SELECT RIGHT 1**, **PIANO** (0).

- C40: volume 40.

10 Keep it as a macro: press **SET**, then **PANEL MEMORY 1**.

- Now **PANEL MEMORY 1** puts C40 into any cell.

11 Put the pattern in the song: go to the last position with **MUTE 1** (POS), press **CONDUCTOR RIGHT 2** (a copy of it now follows it), **MUTE DOWN 1** to go to that copy, and set its pattern to yours with **MUTE 2** (PATT).

12 Save: press **LCDR 3**, then **LCDL 5** (SV.NEW), then **START/STOP** (yes).

- The status line says MODULE SAVED !.

Every control

control	does	with SHIFT
LCDL 1-5 (main)	PLAY, PATTERN, STOP, EDIT, RECORD	
LCDR 1-5 (main)	previous module, next module, Disk Op., SAVE, CLEAR	
MUTE 1-4	POS (UP: the position before), PATT, LEN, BPM	edited pattern, PATT, LEN, speed
MUTE 5, 7, 9, 11	channel 1-4: UP heard, DOWN muted	
MUTE 6, 8, 10, 12	channel 1-4: UP the cursor there, DOWN solo	
MUTE 13-16	SAMPLE, VOLUME, FINETUNE, edit skip up/down	REPEAT, REPLEN, metronome speed, channel
PAGE UP/DOWN	16 rows (Disk Op.: 16 modules; HELP: pages)	
DISPLAY HOLD	SHIFT	
EXIT	briefly: back (page, macros, edit mode); held 1.5 s: the end	
HELP	five pages of help (PAGE UP/DOWN, the wheel); again or EXIT: back	
OTHER PARTS/TG	quadrascope or spectrum analyzer	
TEMPO/PROGRAM	the row; while playing, the tempo; in the Disk Op., the module	the value under the cursor
TRANSPOSE -/+	the note a semitone	the track a semitone
R1/R2 OCTAVE -/+	the note an octave	the track an octave
PART SELECT	LEFT: cursor left; RIGHT 1: cursor right; RIGHT 2: next channel	
CONDUCTOR	LEFT/RIGHT 1: the edited pattern; RIGHT 2: insert a position	RIGHT 2: delete the position
SOLO	solo the cursor's channel (again: all heard)	
SOUND GROUP	hex digits 0-7 (top), 8-F (bottom)	
SOUND EXPLORER	delete the note or the effect	delete the cell
EW EXPANSION	delete the cell	
RHYTHM GROUP	samples 1-16	samples 17-31
AUTO PLAY CHORD	OFF/ON: chord entry; MODE: MULTI	
MULTI	MULTI	
SOUND ARRANGER	OFF/ON: metronome; SET: on the cursor's channel	
ONE TOUCH PLAY	play one row and move on	
FADE IN / OUT	the sample's volume up / down	
TAP TEMPO	the tempo from taps, a tap a beat	
SYNCHRO & BREAK	play the pattern over and over	
VARIATION 1-4	row 0, 16, 32, 48	store the current row there
FILL IN 1 / 2	insert / delete a row in the track	in the pattern
INTRO & ENDING 1 / 2	the song's start / row 63	
START/STOP	play the song, or stop; a question: yes	
PADS 1-6	cut, copy, paste the track (1-3) and the pattern (4-6)	
PADS BANK / STOP	undo / stop	
SEQUENCER PLAY / EASY REC	play the song / the edit mode (while playing: record)	

(Continued on the next page)

control	does	with SHIFT
DISK, SD	the Disk Op.	
CUSTOMIZE	options: blank zeroes, dots, sharps or flats, velocity as Cxx, the scope	
PANEL MEMORY 1-8	the effect macro into the cell (after SET: store)	
SET	store a macro (then 1-8); a question: yes	
BANK VIEW	show the macros	
keybed	notes (C2 = C-1); chords with chord entry	
sustain pedal	a rest	
modulation wheel	the effect parameter under the cursor	

Controls left to the instrument

The volume sliders and **CONTRAST** work as always. These do nothing while ProTracker runs: **DEMO**, **MUSIC STYLIST**, **MUSIC STYLE ARRANGER**, **SPLIT POINT**, **TECHNI-CHORD**, **PROGRAM MENUS**, **CUSTOM PANEL**, **FAVORITES**, **NEXT BANK**, **PADS AUTO SETTING**, the **PART** and **GLOBAL EFFECT** buttons (but **MULTI**), and the pitch-bend wheel.

- While ProTracker runs, the instrument's own functions behind these buttons are kept from the instrument: pressing **PANEL MEMORY 1** puts an effect into a cell and does not recall a registration.

ProTracker's effects

An effect is a command digit and a two-digit parameter (x and y are its digits, xx both). Effects work on the channel they are in, at that row.

effect	what it does
0xy	arpeggio: the note, +x, +y semitones in turn
1xx	slide the pitch up
2xx	slide the pitch down
3xx	slide to the note (portamento)
4xy	vibrato: speed x, depth y
5xy	3 (go on) and volume slide
6xy	4 (go on) and volume slide
7xy	tremolo: speed x, depth y
9xx	start the sample xx00 bytes in
Axy	volume slide: up x or down y
Bxx	jump to position xx
Cxx	volume xx (00-40)
Dxy	to the next position, row xy (decimal)
Fxx	speed (01-1F) or tempo in BPM (20-FF)

effect	what it does
E0x	the Amiga's filter (not heard here)
E1x	fine slide up
E2x	fine slide down
E3x	glissando on/off
E4x	vibrato waveform
E5x	finetune
E6x	loop: E60 marks, E6x plays x times
E7x	tremolo waveform
E9x	retrigger every x ticks
EAx	fine volume up
EBx	fine volume down
ECx	cut the note after x ticks
EDx	delay the note x ticks
EEx	delay the pattern x rows
EFx	invert loop (not heard here)

What differs from the Amiga

- The pattern shows 13 rows (the display has 240 lines); the VU meters are a little shorter.
- The sound is mono and unfiltered: the Amiga's left/right channels and its filters (E0x) are not there.
- A sample loop that starts on an odd word may begin up to three samples early, once.
- A loop changed while a sample plays takes over up to a tick late.
- EFx changes only ProTracker's copy of the sample, not what the DSP plays.
- There is no sample editor, no sampling, no MOD2WAV, and packed (PowerPacker, XPK) modules do not load.
- Modules are listed by their 8.3 names, not their long names.

Troubleshooting

display	what to do
CAN'T FIND FILE !	The module could not be read from the card: check that the card is in, and that the file is in PROTRACK\.
NOT A MOD FILE ! UNSUPPORTED MOD !	Not a 4-channel module, or a packed one.
NO DSP: SILENT ! DSP NOT READY !	The effects DSP could not be used: the program runs silent. Leave and start again.
NOT IN EDIT MODE	Press **EASY REC** first.
MODULE TOO BIG !	The module does not fit in memory for saving (more than 376 KB).
LIST IS FULL !	PROTRACK\ has 128 modules, or PTNEW99.MOD exists: move some off the card.
a button does nothing	Check the page (the tags beside the display) and whether SHIFT is held; **HELP** lists everything.

Limits

item	limit
module file	up to 376 KB
samples	up to 268 KB in all; 31 samples
channels	4 (ProTracker, NoiseTracker and Soundtracker modules)
modules in PROTRACK\	128 listed
effect macros	8 (**PANEL MEMORY 1**-**8**)